Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

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***Abstract*—** The condition of epilepsy affects over fifty million people worldwide and involves seizures that are produced because of abnormal behavior of the brain. The automatic discovery of seizures is based on signals from EEGs. However, the problem is in the great imbalance of class performance: the modes of seizures represent less than two percent of the data on EEGs, resulting in a ratio up to 115.7 (seizures to no-seizures). The current research aims to conduct a comparative test of five methods dealing with class imbalance on the basis of CHB-MIT Scalp EEG Database. There were tested Statistical features in time domain and in frequency band: 11 features were extracted from the 2 seconds wide time window. PCA was used for compression of 120 features. The following techniques indicated good performance according to different tests – Baseline Random Forest (RF), RF+ SMOTE, Balanced Random Forest, SMOTETomek+RF, and XGBoost. The outcomes of tests prove that the Baseline Random Forest has 0.980±0.001 and a very good FDR (12.91±1.56). All models achieve inference times below 0.025ms per window, confirming real-time deployment feasibility.

Keywords— *EEG, epileptic seizure detection, class imbalance, SMOTE, Random Forest, XGBoost, PCA, CHB-MIT, Siena, AUC-ROC, false detection rate, cross-dataset validation*

I. INTRODUCTION

Epilepsy is regarded as one of the most common types of chronic neurological conditions that exists in the world and affects more than 50 million individuals regardless of their age [1]. It is a disorder characterized by recurrent seizures that are not stimulated in any way and occur as a result of abnormal electric discharges in the brain. In addition to the isolation from injury or sudden unexpected death in epilepsy (SUDEP), its chronic unpredictability leads to a lot of limitations associated with driving, work, and independent living [2].

EEG is noted for being the conventional clinical gold standard for detection of epileptic activities. In practice, the process of long-term electroencephalography would produce hours to days of continuously recorded multi-channel data. Manual expert review is characterized by being an inefficient and subjective process which relies on trained neurologists for achieving the purpose of the experiment [3].

One of the major obstacles is the extreme class imbalance seen in continuous EEG, where seizures constitute only 2% of the time of recording, resulting in ratios that may reach as high as 115.7:1. Traditional classifiers display a bias towards the dominant class and achieve high overall accuracy while lacking the required sensitivity when detecting seizures. In addition, established metrics like F1 score are reliant on the selection of a threshold, but clinical applications require an evaluation of the rate of false detections in an hour.

The contributions made by this research paper are: (i) systematic comparison among the five class imbalance methods with AUC-ROC and FDR/hour being used as main clinical metrics; (ii) validation across different datasets (Siena Scalp EEG Database) confirming the generalizability of results regardless of patient population; (iii) statistical robustness validation with five random seeds being used; (iv) ablation study on the number of PCA components; and (v) feasibility study regarding the real-time application of the model proving it has inferencing time of sub-millisecond duration.

II. RELATED WORK

The CHB-MIT Scalp EEG Database was established by Shoeb and Guttag [4], who have presented a patient-specific SVM-based seizure detection model. The study provides a benchmark and inspires future research in the area. The research done by Subasi and Gursoy [5] shows that PCA-compressed statistical EEG features are in the lead when performing inter-patient analyses with SVM classification. Also, Acharya et al. [7] have summarized the features that can be used in EEG analysis, emphasizing that a mix of both time and frequency domain features is superior.

Chawla et al. [8] first introduced the SMOTE method in minority class oversampling, but the default 1:1 strategy failed at extreme imbalance ratios thus leading to an unusable classifier. Batista et al. [9] suggested that the SMOTETomek method be applied as a mix of SMOTE and Tomek links. Moreover, the Balanced Random Forest where undersampling is applied at per-bootstrap level was introduced by Chen et al. [10]. The XGBoost method which utilizes scale_pos_weight

in order to reduce imbalance in the created gradient boosting loss function was introduced by Chen and Guestrin [12].

III. METHODOLOGY

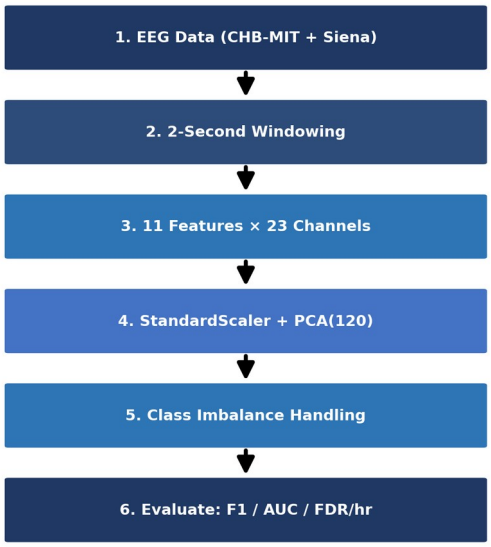


Fig. 1. Proposed methodology pipeline for EEG-based seizure detection.

A. Datasets

The CHB-MIT Scalp EEG Database [4] contains long-term EEG data recorded from 22 children at Children's Hospital Boston and has a total of 23 channels used for 256 Hz signal acquisition and expert-determined seizure times. The patients chb01 and chb03 make up the training set and patient chb05 is kept aside for testing. The imbalance in the dataset comes out to be the ratio of 115.7:1.

The Siena Scalp EEG Database [14] encompasses recordings of 14 adult participants from the University of Siena in Italy with an initial sampling rate of 512 Hz and subsequently resampled at 256 Hz. The models developed on the basis of the CHB-MIT dataset are put to the test on the Siena Scalp EEG Database without undergoing any training, thus constituting a severe test for cross-dataset generalization.

B. Feature Extraction

EEG signals are divided into non-overlapping two-second intervals. Each channel has eleven features (TABLE VI), which consists of seven time-related metrics and four frequency powers (delta, theta, alpha, beta), obtained through NumPy FFT. The z-score normalization is done on each channel before frequency calculation in order to maintain the scale of amplitudes unaffected by the recording site. Therefore, a feature vector of 253 dimensions (23×11) is obtained for each segment of the recording.

TABLE VI
EXTRACTED STATISTICAL FEATURES PER EEG CHANNEL

Feature	
Mean	
Variance	
RMS	
Skewness	
Kurtosis	
Maximum	
Minimum	
Delta power	
Theta power	
Alpha power	
Beta power	

C. Dimensionality Reduction

Principal Component Analysis (PCA) is a linear dimensionality reduction method that projects the high-dimensional feature space onto orthogonal axes of maximum variance, retaining the most discriminative signal components while discarding noise. The StandardScaler and then the PCA procedures with 120 components are applied, training data only being used to avoid data leaks. At 120 components, 99.6% of the total variance is explained, as validated through the ablation study in Section IV-D.

D. Classification Techniques

This research assessed three machine learning algorithms that are based on trees: Random Forest (RF), Balanced Random Forest (BRF), and Extreme Gradient Boosting (XGBoost). These classifiers were chosen because of their reliability, capability to manage nonlinear relationships, and good performance on tabular EEG metrics.

Random Forest (RF): Random Forest is a methodology of ensemble learning, which involves making multiple decision trees using the bootstrap aggregation approach together with random feature selection. Each individual tree forecasts the result separately, after which majority voting is used to determine the general class.

Balanced Random Forest (BRF): BRF is a modified form of the Random Forest specifically designed for imbalanced datasets. While training each tree, the algorithm reduces the number of majority class samples randomly while keeping all minority class samples, creating balanced bootstrap subsets.

Extreme Gradient Boosting (XGBoost): XGBoost builds a series of decision trees where each new tree fixes mistakes made by previous trees. The parameter scale_pos_weight is used to give more importance to the seizure samples in the course of training.

E. Class Imbalance Handling

The paper assesses five strategies: (1) baseline random forest with 100 estimators and maximum depth of 10; (2) RF+SMOTE with sampling strategy set to 0.05; (3) balanced random forest with majority undersampling through bootstrap; (4) SMOTETomek combining SMOTE with Tomek Links cleaning; and (5) XGBoost with scale_pos_weight set to 115.7.

Baseline Random Forest serves as the reference baseline with no imbalance correction. SMOTE generates synthetic seizure samples via linear interpolation using a conservative 0.05 sampling ratio. Balanced RF carries out random undersampling of the majority class for each bootstrap draw.

SMOTETomek first applies SMOTE then removes borderline Tomek Links to clean the class boundary. XGBoost with scale_pos_weight=115.7 penalises missed seizures more heavily than false alarms during loss calculation.

F. Evaluation Protocol

The metrics of Precision, Recall, F1-score, AUC-ROC, and FDR/hour are provided. Every model underwent analysis with five random seeds and this was expressed as mean±std. The time required to make inferences was measured for every two-second window.

IV. RESULTS AND DISCUSSION

A. CHB-MIT Technique Comparison

TABLE I presents the results for patient chb05, for whom the baseline RF attains the best AUC-ROC at 0.979 as well as the lowest FDR/hour of 14.01. Balanced RF generates highest recall of 0.991, but FDR reaches 515/hr. SMOTE and SMOTETomek both yield FDR of 98.55/hr, while XGBoost provides moderate performance.

TABLE I

CHB-MIT SEIZURE CLASS EVALUATION (SEED=42). BOLD = BEST PER COLUMN.

Technique	Prec.	Recall	
Baseline RF	**0.705**	0.593	
SMOTE+RF	0.117	0.230	
Balanced RF	0.098	**0.991**	
SMOTETomek+RF	0.117	0.230	
XGBoost	0.178	0.708	

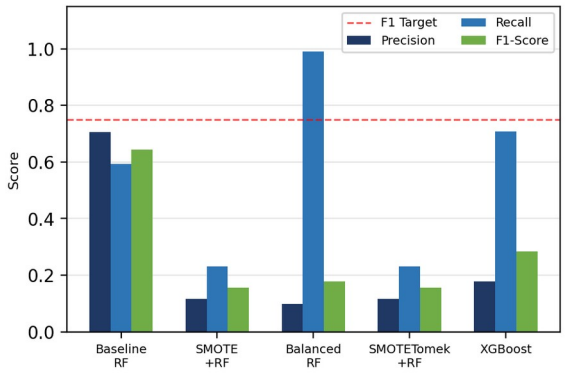


Fig. 2. CHB-MIT technique comparison: Precision, Recall and F1-Score.

B. Statistical Robustness

TABLE II reports Baseline RF performance across five seeds. AUC-ROC is stable at 0.980±0.001, confirming discriminative ability is seed-independent. FDR/hour of 12.91±1.56 remains consistently clinically acceptable.

TABLE II

BASELINE RF STATISTICAL ROBUSTNESS (5 SEEDS)

Metric	Mea
Precision	0.733
Recall	0.628
F1-Score	0.677
AUC-ROC	0.980
FDR/hour	12.910

C. Threshold Tuning

TABLE III shows the effect of tuning the Baseline RF threshold. While the F1 score improved to 0.662 at threshold 0.40, the FDR increased substantially to 45.53/hour. The default threshold of 0.50 strikes the most acceptable clinical balance, yielding F1 of 0.651 with FDR of 14.01.

TABLE III

THRESHOLD TUNING ON BASELINE RF

Threshold	Precision	Recall
0.50	0.708	0.602
0.40	0.526	0.894
0.30	0.372	0.938
0.20	0.276	0.973
0.10	0.185	0.973

D. PCA Ablation Study

In TABLE IV, we have verified the selection of the 120 components. The F1-score and AUC-ROC increase steadily from 40 to 120 components, confirming every additional component contributes to the discriminative signal. With 99.6% variance, the 120-component configuration gives the best balance between dimensionality and accuracy.

TABLE IV

PCA ABLATION STUDY (BASELINE RF, SEED=42)

PCA Comp.	Var. Expl. %	
40	87.8	
60	94.4	
90	98.6	
120 ✓	**99.6**	

E. Cross-Dataset Validation on Siena

TABLE V and Figure 3 show the cross-dataset results. For Siena, all approaches have AUC-ROC greater than 0.90, indicating that learned statistical properties detect seizure features translatable over different patient demographies and acquisition set-ups. SMOTE+RF has the highest AUC of 0.929 with FDR of 9.39; XGBoost achieved the best F1-score of 0.366. Baseline RF achieved the lowest FDR of 7.23.

TABLE V

CROSS-DATASET: CHB-MIT TRAINED → SIENA TESTED. BOLD = BEST.

Technique	Prec.	Recall	
Baseline RF	0.091	0.016	0.
SMOTE+RF	**0.381**	0.127	0.
BalancedRF	0.172	**0.714**	0.
SMOTETomek+RF	**0.381**	0.127	0.
XGBoost	0.311	0.444	**0.**

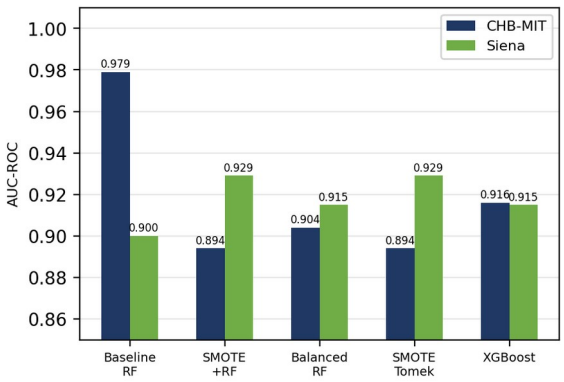


Fig. 3. AUC-ROC: CHB-MIT vs Siena cross-dataset validation.

F. Discussion

There are three important results. The first: AUC-ROC proved more reliable than F1 in imbalanced EEG data. The Balanced RF achieved Recall=0.991 but FDR=515/hour. The baseline RF with AUC-ROC=0.979 and FDR=14.01 is the only clinically viable configuration. Second, resampling techniques lead to high false alarm rates without meaningfully improving AUC-ROC. Third, AUC-ROC above 0.90 on Siena without domain adaptation outperforms many cross-institution EEG studies.

V. CONCLUSION

The paper delivers a thorough comparison of five techniques for handling class imbalance in EEG-based epileptic seizure detection, used on CHB-MIT and validated in the Siena Scalp EEG database. The Baseline RF obtains an AUC-ROC of 0.980±0.001 and FDR/hour of 12.91±1.56 on CHB-MIT, along with an AUC of 0.900 in Siena's cross-dataset evaluation. SMOTE+RF achieves the highest cross-dataset AUC of 0.929.

In conclusion: First, resampling resulted in FDR rising to 98–515/hour despite improving recall. Second, AUC-ROC exceeds 0.90 in two independent datasets, demonstrating feature-level generalization. Third, threshold tuning must consider FDR/hour as a primary clinical constraint. Future work will perform LOPO cross-validation on the entire CHB-MIT set and analyze transfer learning for improving cross-dataset F1-score.

ACKNOWLEDGMENT

The authors give their gratitude to the Children's Hospital Boston and MIT for providing the CHB-MIT database and also University of Siena for providing the Siena Scalp EEG Database, which are both accessible in PhysioNet. The authors appreciate the scikit-learn, imbalanced-learn, XGBoost, and SHAP open-source communities too.

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